

Googling as a Key: Exploring the Relationship between COVID-19 Cases and Internet Search Behavior

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Abstract— In this study, we analyzed the relationship between the search behavior of users on COVID-19-related queries and the rates of COVID-19 cases. To do this, we first cleaned and visualized data from the COVID-19 Data Repository by the Center for Systems Science and Engineering (CSSE) at Johns Hopkins University, and also gathered data on the most frequently searched COVID-19-related queries using Google Trends. We observed a similarity in the search rates for these queries, which led us to investigate their correlation with COVID-19 cases. To investigate this correlation, we superimposed the search behavior data on top of the COVID-19 case data. We then conducted two statistical tests to further analyze our dataset, which helped us gain insights into the relationship between search behavior and COVID-19 cases. We found a significant relationship between the two variables, which has implications for understanding the public's awareness and response to the COVID-19 pandemic. Our findings suggest that monitoring search behavior may be a useful tool in tracking the spread of the disease and informing public health policies.

Keywords— COVID-19, data analysis, Google Trends, search behavior, correlation

I. INTRODUCTION

Infectious diseases have become a serious public health problem in the 21st century [1]. Controlling the spread of such diseases is crucial, and surveillance systems are an essential tool for top management. These systems are responsible for gathering targeted, dependable, and timely evidence, and presenting it effectively and efficiently. Various strategies, including Periodic Population-based Surveys and Laboratory-based Surveillance, have been developed to meet the management needs [2]. However, traditional surveillance data are primarily collected from clinics, hospitals, laboratories, and other health providers, limiting the system's ability to respond quickly to unforeseen changes. This leads to slower and time-consuming processes, hindering the ability of health agencies and governments to respond promptly to unforeseen changes [3]. Effective public health practices rely heavily on surveillance, with the primary goal of contributing to the health of the population by providing valid information that can be used to guide public health interventions promptly [4]. Therefore, alternative methods such as using search engine data have been explored as a complementary approach to traditional surveillance systems for infectious disease surveillance and prediction. In this paper, we describe our attempt to obtain data on COVID-19 cases in Iran and explore the possibility of using Google Trends to predict future

outbreaks. Our goal is to examine the relationship between internet search behavior and disease activity and determine whether we can develop models that effectively forecast future outbreaks.

II. LITERATURE

COVID-19 has been a global health crisis that has affected millions of people worldwide. With the rise of technology, researchers have found new ways to track the spread of the virus and predict outbreaks. In a study led by Amaryllis Mavragani [5], online search data was analyzed to track the spread of the virus in Europe, revealing a significant correlation between online interest and the number of cases and deaths. This shows the potential of using Google Trends data for disease surveillance and preparedness.

Another study by Tichakunda Mangono and colleagues [6] aimed to understand how information-seeking behavior changes during pandemics. By analyzing Google Trends data, the researchers found that changes in information-seeking often occurred before government action, indicating the need for a real-time dashboard for decision-making. Maria Effenberger and her team [7] used Google Trends to track public interest in the virus and found that global interest peaked before the peak of newly infected cases, suggesting that these indices could help monitor the progression of the pandemic.

Jacopo Ciaffi and colleagues [8] found a correlation between Google searches for COVID-19 symptoms and an increase in ICU admissions and deaths. They suggest that using symptom-related search terms could be a better predictor of the outbreak's progression compared to more general search terms. In Cuilian Li's study [9], Google Trends searches for coronavirus and pneumonia were highly correlated with the daily incidence of COVID-19 cases, demonstrating the potential of internet surveillance tools in predicting disease outbreaks.

Finally, Samira Yousefinaghani and her team conducted a comparative study on the potential of Google searches and Twitter activities to be used in early warning systems for COVID-19 in Canada and the US. They found that Google searches for COVID-19 symptoms and precautionary measures had strong correlations with official cases in Canada and performed better in predicting the onset and peak of the first waves.

These studies highlight the potential of using Google Trends data for disease surveillance and early warning systems,

providing valuable insights for policymakers and public health officials in managing future outbreaks.

III. METHODOLOGY:

ANALYZING THE RELATIONSHIP BETWEEN COVID-19 CASES AND INTERNET SEARCH BEHAVIOR USING GOOGLE TRENDS AND STATISTICAL TESTS

Initially, we exported the data from the COVID-19 Data Repository by the Center for Systems Science and Engineering (CSSE) at Johns Hopkins University [10] and subsequently cleaned it to avoid any inaccuracies. These data were used to generate a plot, as shown in Fig. 1.

Subsequently, we used Google Trends [11] to identify the most frequently searched queries from February 19, 2020, to February 12, 2023. Our selection included the following search terms, which were a combination of Persian and English: کرونا (corona in Persian), علائم کرونا (corona symptoms in Persian), ویروس کرونا (coronavirus in Persian), داروی کرونا (corona medicine in Persian), ثبت نام واکسن کرونا (Corona vaccine registration in Persian), and coronavirus (in English).

We also visualized the search rates for selected queries (Fig. 2).

We observed that the search rates for the different keywords were similar during this period. This led us to investigate whether there was a correlation between this search behavior and the rates of COVID-19 cases.

Therefore, we superimposed the two previous graphs on top of each other; Fig. 3 shows the results.

Fig. 3 contains significant insights into search behavior. Before the outbreak, search rates were notably high, as people were uninformed, confused, and anxious. During the first peak at the end of 2020, search rates were particularly close to COVID-19 case rates, indicating a significant correlation between the two. A similar pattern is observed for the second and third peaks. However, around January 2022, a notable change in search behavior was observed. Despite the high COVID-19 case rates, search rates are not as high as expected. This could be due to the increased knowledge and awareness among people.

Overall, the superimposed graph can be a useful tool for researchers and policymakers to better understand how people's search behavior changes in response to the dynamic nature of outbreaks.

Toward the end of our data analysis journey, we conducted two statistical tests to further analyze our dataset. Below, we will provide an explanation of each test and how they helped us to gain insights into our data.

Initially, we conducted a Granger causality test which is a statistical method that can help us understand the relationship between two variables by identifying the direction of causality. This information is crucial for understanding the nature of the relationship between variables, especially when studying complex phenomena like COVID-19 cases and internet search behavior.

Unlike other tests, such as correlation or regression analysis, the Granger causality test can determine not only the strength and nature of the relationship but also the direction of causality. By doing so, it can help us understand which variable

is driving changes in the other, allowing for a more complete understanding of the relationship between COVID-19 cases and internet search behavior.

Another benefit of the Granger causality test is that it is designed for time series data. This means that it is well-suited for analyzing the relationship between COVID-19 cases and internet search behavior over time. Other tests, such as cross-sectional analysis, may not be appropriate for time series data. This makes the Granger causality test particularly useful when studying trends and changes over time.

Finally, the Granger causality test can account for the lagged effects of one variable on another. This is important because changes in one variable may not have an immediate impact on the other, but may take some time to become evident. Other tests may not account for these lagged effects, which can lead to inaccurate results.

we utilized the "grangers_causation_matrix()" function from the statsmodels.tsa.stattools module in Python. The test results are presented in a square pandas data frame, showing the minimum p-values for each pair of time series variables.

We set the maximum lag to 12 and investigated whether changes in Covid cases could predict changes in Google search. Table I presents the results.

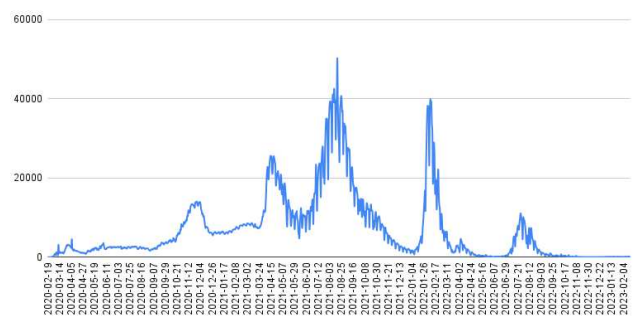


Fig. 1. COVID-19 cases in Iran

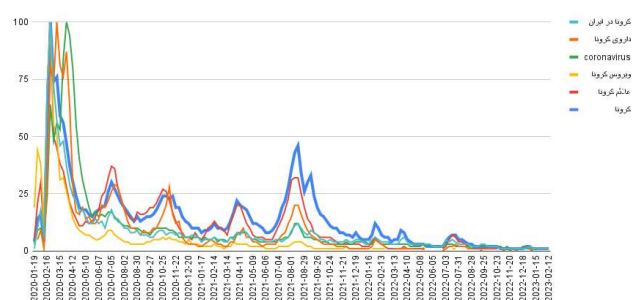


Fig. 2. Most Frequently Searched Queries in Iran from February 19, 2020, to February 12, 2023.

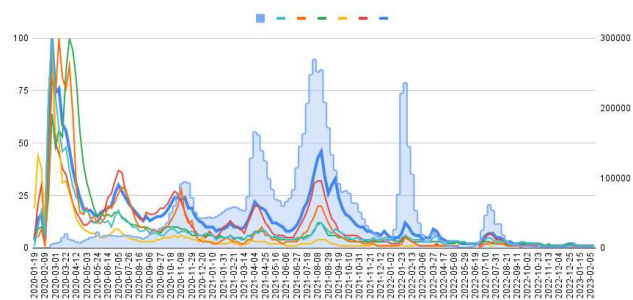


Fig. 3. Superimposition of COVID-19 cases In Iran (Fig. 1.) and Most Frequently Searched Queries in Iran from February 19, 2020, to February 12, 2023 (Fig. 2.).

The results of the test indicated that changes in Covid cases may be causing changes in Google Search, but changes in Google Search are not causing changes in Covid Cases. The coefficient of 0.06 in the first row of the table indicates that there is some evidence to suggest that changes in Data can help predict changes in Google search, while the coefficient of 0.0 in the second row of the table indicates that changes in Google search do not provide any useful information about how Covid Cases will change.

Next, we conducted Johanson's Cointegration test. We selected this test for our study as it is a robust statistical tool that addresses the issue of non-stationarity in time series data and is capable of analyzing multivariate time series. Compared to other cointegration tests, such as the Engle-Granger and Phillips-Ouliaris tests, Johansen's cointegration test provides several benefits. It is specifically designed to analyze multivariate time series, making it an appropriate choice for investigating complex relationships between multiple variables. Additionally, it can estimate the number of cointegrating vectors between time series, enabling us to determine the existence and number of long-term relationships between the variables. The test also yields more reliable results by accounting for multiple cointegrating vectors. Lastly, Johansen's cointegration test offers the flexibility of lag selection, enabling us to choose the appropriate lag length for our analysis.

We performed this test using the `coint_johansen` function from the `statsmodels.tsa.vector_ar.vecm` module in Python.

To run the test, we used the `cointegration_test` function on a pandas data frame called 'df'. This function takes an optional argument 'alpha', which determines the level of significance for the test. The results of the test are presented in Table II which includes the name of each variable, the test statistic, and the significance level.

The first column lists the names of the variables being tested. The second column shows the test statistic, which is a measure of how strong the evidence is for the relationship between the variables. The third column shows the critical value, which is the value at which we can be confident that there is a relationship between the variables.

In our case, the critical value was set to C (95%), which means that we are 95% confident that there is a relationship between the variables if the test statistic is greater than a certain value.

The fourth column of the summary shows whether the test statistic is greater than the critical value. If it is, then we can say that the relationship between the variables is statistically significant.

TABLE I. GRANGER CAUSALITY TEST RESULT

	Covid Cases_x	Google Search_x
Covid Cases_y	1.0	0.06
Google Search_y	0.0	1.0

TABLE II. JOHANSON'S COINTEGRATION TEST RESULTS

Name	Test Stat	>	C (95%)	=>	Signif
Covid Cases	49.49	>	12.3212	=>	true
Google Search	4.15	>	4.1296	=>	true

In our case, both variables have a test statistic that is greater than their respective critical values, which means that there is a statistically significant long-term relationship between "Covid cases" and "Google Search".

IV. CHALLENGES OF MODELING WEB SEARCH BEHAVIOR FOR ACCURATE OUTBREAK PREDICTION

One major challenge that we encountered, particularly in Iran, was the issue of internet censorship. This resulted in users resorting to virtual private networks (VPNs) and changing their IP addresses, which can impact their geographical location and reduce the accuracy of the data we could obtain from Iranian users.

Internet penetration is another important metric. Despite the high Internet penetration rates in some countries, there are still difficulties in predicting future search behavior based on the retrospective nature of query searching. Special keywords represent the search behavior of individuals during a particular timeframe. Although they provide valuable insights, they cannot guarantee accurate and consistent predictions of future search behavior. To build accurate predictive models, long-term data support is necessary for time-series analysis [12].

In the case of the seasonal flu, for example, the timing of its onset varies from one season to another. Thus, it is essential to quickly detect the start of the flu season to control the spread of the virus and minimize the impact of the disease.

Health officials can prepare preventive strategies once the flu season begins. These strategies include reinforcing influenza prevention messages and expanding vaccination campaigns to protect the public.

Another behavioral challenge observed in web searches is that awareness levels change over time [13]. As people become more familiar with a disease, they tend to search less frequently for information related to their symptoms (Fig. 3). This change in search behavior poses a challenge in predicting the onset of a disease outbreak, emphasizing the need for continuous monitoring of search behavior to understand how public awareness of the disease evolves over time.

V. CONCLUSION AND FUTURE DIRECTIONS FOR INFECTIOUS DISEASES SURVEILLANCE AND PREDICTION

After conducting both the Granger test and Johansen test, we conclude that there is indeed a significant relationship between the growth of COVID-19 cases and the search behavior of internet users for COVID-19-related keywords. The Granger test revealed that the search behavior of internet users for COVID-19-related keywords can be a valuable predictor of the growth of COVID-19 cases.

The results of the Johansen test indicated cointegration between the search behavior of internet users for COVID-19-related keywords and the growth of COVID-19 cases. This implies a long-term relationship between these two variables.

Furthermore, In light of the complexity of the search behavior discussed in the previous section, future research can benefit from building models that aim to improve the accuracy of predictions. In addition, early detection can assist health administrators in making optimal staffing and resource allocation decisions to prepare for potential surges in patient visits to hospitals.

The earlier we detect a disease onset, the more time we have to plan and implement proactive prevention strategies to stop its spread. It should be noted that this preparation method is not easy or quick.

In summary, while modeling web search behavior can improve the accuracy of outbreak predictions, it is equally crucial to remember that preparing for potential surges in patient visits is not a quick and easy task. However, early detection remains an essential tool for health administrators to plan and implement proactive prevention strategies to control disease spread.

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